

Foundations of Data Engineering and Data Modeling

Introduction to Data Engineering

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Learning Objectives

After completing this lecture, students will be able to:

- Explain the role and responsibilities of a Data Engineer
- Describe end-to-end data engineering workflows
- Understand the modern data stack architecture
- Differentiate between OLTP and OLAP systems
- Explain data flow from transactional to analytical systems

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1 What is Data Engineering?

Data Engineering is the discipline focused on designing, building, and maintaining systems that collect, store, process, and deliver data for analytics, reporting, and decision-making.

Key Motivation:

- Data is generated at high volume and velocity
- Raw data is often inconsistent and unstructured
- Analytics and machine learning require clean, reliable data

1.1 Role of a Data Engineer

A Data Engineer is responsible for:

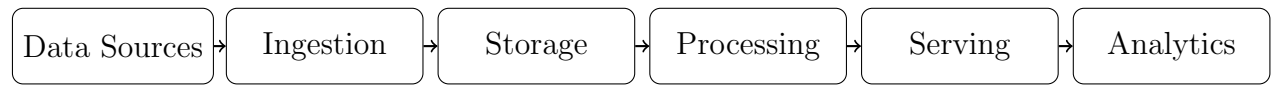
- Designing data ingestion pipelines
- Implementing ETL / ELT workflows
- Managing data storage systems
- Ensuring data quality and reliability
- Supporting analytics and machine learning workloads

1.2 Comparison with Related Roles

Role	Primary Focus	Output
Data Engineer	Pipelines and infrastructure	Clean, reliable data
Data Scientist	Modeling and prediction	Insights, ML models
Data Analyst	Reporting and visualization	Dashboards, reports

2 End-to-End Workflow

A typical data engineering workflow is shown below:



2.1 Workflow Stages

Data Sources

- Transactional databases
- APIs
- IoT sensors
- Log files

Data Ingestion

- Batch ingestion (periodic data loads)
- Streaming ingestion (real-time data)

Data Storage

- Data Lakes for raw data
- Data Warehouses for structured, processed data

Data Processing

- Data cleaning
- Transformation and aggregation
- Feature engineering

Data Consumption

- Dashboards
- Reports
- Machine learning models

3 Evolution of Data Platforms

Traditional data platforms were monolithic and on-premise, whereas modern data platforms are cloud-native, scalable, and modular.

3.1 Layers of the Modern Data Stack

Layer	Purpose	Examples
Data Sources	Generate data	Applications, sensors
Ingestion	Collect data	Kafka, APIs
Storage	Persist data	S3, BigQuery
Processing	Transform data	Spark, dbt
Orchestration	Schedule workflows	Airflow
Analytics	Visualization	Power BI, Tableau

3.2 Advantages of Modern Data Stack

- Scalability
- Cost efficiency
- Faster development cycles
- Separation of compute and storage

3.3 OLTP Systems

OLTP (Online Transaction Processing) systems are designed to support day-to-day business operations.

Characteristics:

- High number of short transactions
- Frequent INSERT, UPDATE, DELETE operations
- Normalized database schemas

Examples:

- Banking systems
- E-commerce platforms
- Student registration systems

3.4 OLAP Systems

OLAP (Online Analytical Processing) systems support analytical queries and decision-making.

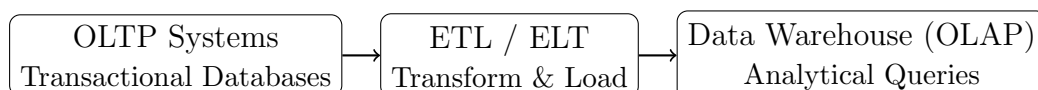
Characteristics:

- Complex read-heavy queries
- Historical data analysis
- Denormalized schemas (Star/Snowflake)

3.5 OLTP vs OLAP Comparison

Feature	OLTP	OLAP
Purpose	Operations	Analysis
Transactions	Short	Complex
Data	Current	Historical
Schema	Normalized	Denormalized
Users	End users	Analysts

3.6 Data Flow Relationship



Summary

- Data Engineering enables reliable analytics
- Modern workflows handle scale and complexity
- Modern data stacks are cloud-native
- OLTP and OLAP systems serve complementary purposes

4 Suggested Reading

1. Martin Kleppmann, *Designing Data-Intensive Applications*, O'Reilly Media.
2. Joe Reis and Matt Housley, *Fundamentals of Data Engineering*, O'Reilly Media.
3. Ralph Kimball, *The Data Warehouse Toolkit*, Wiley.
4. **Apache Spark Documentation:** <https://spark.apache.org>
5. **Apache Kafka Documentation:** <https://kafka.apache.org>
6. Google Cloud Data Engineering Documentation

5 Suggested Questions

1. What are the main responsibilities of a Data Engineer?
2. Explain the difference between OLTP and OLAP with examples.
3. Describe the layers of a modern data stack.
4. Why is data transformation important in data engineering?

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1 Data Types and File Formats in Data Engineering

Data engineering systems deal with diverse data types and file formats depending on the source, processing requirements, and analytical goals. Understanding data types and choosing the appropriate file format is critical for building efficient, scalable, and reliable data pipelines.

1.1 Data Types in Data Engineering

Data types define how data is represented, stored, and processed in systems.

Structured Data

Structured data follows a predefined schema with fixed fields and data types.

Examples:

- Relational database tables
- CSV files
- Data warehouse fact and dimension tables

Characteristics:

- Tabular format (rows and columns)
- Strong schema enforcement
- Easy to query using SQL

Semi-Structured Data

Semi-structured data does not follow a rigid schema but contains tags or keys to describe the data structure.

Examples:

- JSON documents
- XML files
- Avro records

Characteristics:

- Flexible schema
- Supports nested and hierarchical structures
- Common in APIs and event-based systems

Unstructured Data

Unstructured data lacks any predefined format or schema.

Examples:

- Text documents
- Images, audio, video
- Social media posts

Characteristics:

- Difficult to analyze using traditional SQL
- Requires preprocessing and feature extraction
- Often used in big data and AI applications

1.2 File Formats in Data Engineering

File formats define how data is physically stored on disk or transmitted between systems.

1.3 CSV (Comma-Separated Values)

CSV is a simple, text-based format used to store tabular data.

Key Features:

- Plain text format
- One record per line
- Values separated by delimiters (comma, tab)

Use Cases:

- Data exchange with Excel and databases
- Small to medium datasets
- Initial data ingestion

Limitations:

- No schema enforcement
- No support for nested data
- Inefficient for large-scale analytics

1.4 JSON (JavaScript Object Notation)

JSON is a lightweight, semi-structured data format widely used in web systems.

Key Features:

- Key-value pairs
- Supports nesting and arrays
- Human-readable

Use Cases:

- REST APIs
- Configuration files
- NoSQL databases

Limitations:

- Larger file size
- Slower parsing
- Not optimized for analytical workloads

1.5 Avro

Apache Avro is a binary, schema-based serialization format designed for big data systems.

Key Features:

- Explicit schema definition
- Supports schema evolution
- Compact binary storage

Use Cases:

- Kafka message serialization
- Distributed data pipelines
- Long-term data storage in HDFS

Limitations:

- Not human-readable
- Requires schema management

1.6 Parquet

Apache Parquet is a columnar storage format optimized for analytical workloads.

Key Features:

- Column-oriented storage
- Efficient compression and encoding
- Optimized for OLAP queries

Use Cases:

- Data warehouses
- Spark and Hive analytics
- Machine learning pipelines

Advantages:

- Fast query performance
- Reduced I/O
- Efficient storage for large datasets

1.7 Comparison of File Formats

Feature	CSV	JSON	Avro	Parquet
Data type	Structured	Semi-structured	Semi-structured	Structured
Human readable	Yes	Yes	No	No
Schema support	No	Implicit	Explicit	Explicit
Nested data	No	Yes	Yes	Limited
Storage format	Row-based	Row-based	Row-based	Column-based
Compression	No	No	Yes	Yes
Analytics performance	Low	Low	Medium	High
Big data suitability	Low	Medium	High	Very High

1.8 Practical Recommendation

- Use **CSV** for simple data exchange and small datasets
- Use **JSON** for APIs and flexible data structures
- Use **Avro** for data ingestion and streaming pipelines
- Use **Parquet** for analytical queries and data warehousing

2 When to Use CSV vs JSON

2.1 When to Use CSV

CSV (Comma-Separated Values) format is most suitable when the data is **tabular, flat, and structured**.

Best suited for:

- Structured datasets with rows and columns
- Large datasets where file size and speed are important
- Data exchange with tools such as Excel, R, SQL, and Pandas
- Logs, reports, and simple datasets without nested relationships

Advantages:

- Lightweight and compact format
- Faster read and write performance
- Widely supported across platforms
- Easy to view and edit in spreadsheet applications

Limitations:

- Does not support nested or hierarchical data
- Lacks explicit data type information
- Cannot store complex objects or arrays

Example:

```
student_id,name,department,marks,passed
1,Amit,CSE,85.5,True
```

2.2 When to Use JSON

JSON (JavaScript Object Notation) is preferred when the data is **hierarchical, semi-structured, or nested**.

Best suited for:

- Web APIs and RESTful services
- Configuration files
- Data exchange between applications
- Data containing nested objects or variable structures

Advantages:

- Supports nested objects and arrays
- Self-describing structure using key-value pairs
- Preserves data hierarchy
- Widely used in web and microservice architectures

Limitations:

- Larger file size compared to CSV
- Slower parsing for very large datasets
- Less convenient for spreadsheet-based analysis

Example:

```
{  
  "student_id": 1,  
  "name": "Amit",  
  "department": "CSE",  
  "marks": 85.5,  
  "passed": true  
}
```

2.3 Quick Comparison

Scenario	Recommended Format
Simple tabular data	CSV
Large datasets	CSV
Excel / R / SQL analysis	CSV
Web APIs and applications	JSON
Nested or hierarchical data	JSON
Configuration files	JSON

2.4 Practical Recommendation

In practice, CSV is commonly used for data analysis, reporting, and machine learning pipelines, whereas JSON is preferred for data exchange, APIs, and NoSQL-based systems.

3 Why Avro is Preferred over JSON and CSV in Big Data Systems

In large-scale data engineering and big data ecosystems, data formats must support **high performance, schema evolution, compact storage, and distributed processing**. While CSV and JSON are simple and human-readable, they fall short when data volume, velocity, and variety increase significantly. Apache Avro addresses these challenges and is therefore widely adopted in modern big data pipelines.

3.1 What is Apache Avro?

Apache Avro is a **binary, schema-based data serialization format** designed for efficient data storage and data exchange in distributed systems.

Avro is commonly used with big data frameworks such as:

- Apache Hadoop

- Apache Spark
- Apache Kafka
- Apache Flink

3.2 Key Features of Avro

1. Schema-Based Serialization

Avro stores the data schema separately from the actual data. The schema is written in JSON format and defines:

- Field names
- Data types
- Default values

This ensures:

- Strong data consistency
- Explicit data types
- Easier data validation

Schema Evolution Support

One of Avro's most important advantages is **schema evolution**. It allows schemas to change over time without breaking existing systems.

Supported schema changes include:

- Adding new fields with default values
- Removing unused fields
- Renaming fields

This is critical in long-running data pipelines where data structures evolve.

Compact Binary Format

Unlike CSV and JSON, Avro uses a **binary encoding**, which results in:

- Smaller file sizes
- Reduced network bandwidth usage
- Faster disk I/O

This makes Avro highly suitable for petabyte-scale datasets.

Faster Serialization and Deserialization

Avro is optimized for machine processing rather than human readability. As a result:

- Parsing is significantly faster than JSON
- Data processing jobs run more efficiently

Language and Platform Independence

Avro supports multiple programming languages including:

- Java
- Python
- Scala
- C++

This makes Avro ideal for heterogeneous big data environments.

3.3 Comparison: CSV vs JSON vs Avro

Feature	CSV	JSON	Avro
Human readable	Yes	Yes	No
Schema support	No	Implicit	Yes (Explicit)
Nested data	No	Yes	Yes
File size	Small	Medium	Very Small
Parsing speed	Fast	Slow	Very Fast
Schema evolution	No	Limited	Yes
Big data readiness	Low	Medium	High

3.4 Why Big Data Systems Prefer Avro

Big data systems prioritize:

- Scalability
- Performance
- Reliability
- Schema management

Avro satisfies these requirements by:

- Enforcing structured schemas
- Supporting schema evolution
- Reducing storage and network costs
- Integrating seamlessly with distributed processing frameworks

3.5 Typical Use Cases of Avro

- Data ingestion pipelines
- Kafka message serialization
- Hadoop Distributed File System (HDFS) storage
- Spark-based ETL jobs
- Long-term archival of structured and semi-structured data

3.6 Examples of Data Types and File Formats

This subsection illustrates practical examples of how different data types and file formats appear in real-world data engineering scenarios.

Example 1: Structured Data using CSV

A CSV file representing student examination records:

student_marks.csv

```
student_id,name,department,marks,passed
101,Amit,CSE,85.5,True
102,Riya,EE,72.0,True
103,Rahul,ME,38.5,False
```

Observation:

- Data is strictly tabular
- No nested structures
- Suitable for Excel, SQL, and Pandas analysis

Example 2: Semi-Structured Data using JSON

A JSON document returned by a web API for student information:

```
{
  "student_id": 101,
  "name": "Amit",
  "department": "CSE",
  "courses": ["DBMS", "Big Data", "AI"],
  "address": {
    "city": "Patiala",
    "state": "Punjab"
  }
}
```

Observation:

- Supports nested objects and arrays
- Flexible schema
- Common in REST APIs and NoSQL systems

Example 3: Avro Schema and Record

Avro Schema (student.avsc)

```
{
  "type": "record",
  "name": "Student",
  "fields": [
    {"name": "student_id", "type": "int"},
    {"name": "name", "type": "string"},
    {"name": "department", "type": "string"},
    {"name": "marks", "type": "float"}
  ]
}
```

Observation:

- Explicit schema definition
- Enforces data types
- Schema can evolve over time

(The actual Avro data is stored in compact binary format and is not human-readable.)

Example 4: Parquet Data for Analytics

A Parquet file storing the same student data column-wise:

Logical Representation:

```
Column: student_id -> [101, 102, 103]
Column: name       -> ["Amit", "Riya", "Rahul"]
Column: department -> ["CSE", "EE", "ME"]
Column: marks      -> [85.5, 72.0, 38.5]
```

Observation:

- Column-oriented storage
- Only required columns are read during queries
- Highly efficient for aggregation and filtering

3.7 Real-World Pipeline Example

A typical data engineering pipeline using multiple formats:

1. Data generated by applications is sent as **JSON** via REST APIs
2. Streaming events are serialized using **Avro** in Apache Kafka
3. Raw data is stored in HDFS or cloud storage
4. Transformed analytical data is stored in **Parquet** format
5. Business intelligence tools query Parquet data for reporting

3.8 Practical Recommendation

CSV and JSON are ideal for small-scale data exchange and human interaction. However, for large-scale, production-grade data engineering pipelines, **Avro is preferred due to its performance, schema evolution, and scalability advantages.**

- Use **CSV** for simple data exchange and small datasets
- Use **JSON** for APIs and flexible data structures
- Use **Avro** for data ingestion and streaming pipelines
- Use **Parquet** for analytical queries and data warehousing

3.9 Key Takeaway

Different file formats serve different purposes:

- **CSV** focuses on simplicity
- **JSON** focuses on flexibility
- **Avro** focuses on schema consistency and data exchange
- **Parquet** focuses on analytical performance

Data Ingestion: Concepts, Benefits, Challenges, and Tools

Data Ingestion: Concepts, Benefits, Challenges, and Tools

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1 Introduction

In this chapter, we will discuss *what data ingestion* is, its **benefits** and **challenges**, and the best data ingestion **tools**.

When preparing a report or analysis, a large amount of data must often be reviewed before finalizing the document. This data usually comes from multiple sources and in different formats, making it difficult to gather and analyze efficiently.

Instead of manually collecting and storing information in a data warehouse, organizations can use data ingestion tools. These tools provide features that help create accurate, up-to-date, and well-organized reports by automating the collection and transfer of data.

2 What Is Data Ingestion?

Data ingestion is the process of collecting and importing data from various sources into a storage system or data platform for further processing and analysis. It supports both batch and real-time formats, enabling organizations to work with both historical and up-to-date data.

Data ingestion in big data is also the first step in creating data pipelines used for analytics, where you collect, transform, and load for insights. This involves the initial transfer of data and ongoing updates to ensure the data is current and accurate.

In addition, understanding the data ingestion process flow is crucial because it enables real-time or near-real-time data processing, supports machine learning and AI applications, and drives informed decision-making across the organization.

Regarding data ingestion methods, the two standard ones you can come across are batch and real-time data ingestion. However, micro-batching is another method that has become popular recently. Here is the exact meaning behind these data ingestion methods:

1. **Batch data processing** The data is imported in batches at a regular schedule or interval. For example, a company can have batch data ingestion once per day, enough for their daily reports to be created.
2. **Real-time data processing** The data is imported as it is created or emitted by the source. That means data could be added to the warehouse constantly and streamed as the company or customer needs.
3. **Micro batching** The data is imported in small batches, which are more frequent than the batches from the regular batch data processing method. This is the processing method used by most streaming systems.

2.1 Data Ingestion Methods

The three most common data ingestion methods are:

- **Batch Data Processing** Data is imported in batches at scheduled intervals, such as once per day for daily reports.
- **Real-Time Data Processing** Data is imported continuously as it is generated by the source system.
- **Micro-Batching** Data is imported in small, frequent batches, commonly used by streaming systems.

3 Types of Data Ingestion

3.1 Batch Data Processing

Batch data processing collects and processes data at scheduled intervals. It is useful when real-time access is not required. For example, a retail company may process sales data nightly to update inventory and sales reports.

3.2 Streaming Data Ingestion

Real-time data processing collects and transfers data from source systems as it occurs. That's essential for time-sensitive use cases like stock market trading or power grid monitoring.

For instance, a financial trading platform must process transaction data in real-time to provide up-to-date market information.

3.3 Lambda Architecture

Lambda architecture mixes batch and real-time data ingestion. It enables organizations to process data both in real-time and in batches, delivering a comprehensive approach to data management.

This method is beneficial when you require immediate insights alongside long-term trend analysis.

An example could be an online streaming service using real-time data to recommend content to users while analyzing viewing patterns.

4 Benefits of Data Ingestion

Every business can benefit from data ingestion as all the data they gather will ultimately help them understand the market, their customers, what type of products they need to create, how to improve their company, etc. Below are 5 most common benefits you can get if you start using data ingestion.

Data ingestion provides several advantages:

4.1 Data Availability

Data ingestion stops data silos and allows a more holistic view of organizational data. Data ingestion also ensures all relevant data is readily available for analysis and decision-making by integrating data from various sources into a centralized repository,

For example, your company can combine sales, marketing, and customer service data to get a complete picture of customer behavior.

4.2 Data Insights

Reliable data ingestion offers better decision-making. It provides accurate and up-to-date data, helping you gain valuable insights, identify trends, and make informed decisions.

For instance, if you have an e-commerce company you can analyze customer purchase patterns to optimize its inventory and marketing strategies.

4.3 Automated Data Transfer

Instead of transferring data gathered from other companies or reports yourself, you can use a data ingestion tool to extract, transfer, and store the information. That means you will have much more time to finish other, more important tasks or focus on different aspects of your business to improve it further..

4.4 Extraction of Value from Data

When looking at data from other companies or your own business, data ingestion can help you extract valuable information you can use to improve the business further. This data can help you gain insight into how successful companies work or how you can use gaps in the market to your advantage.

4.5 Data Uniformity

No matter what kind of data you give the data ingestion tool, it can extract the needed information and create a unified dataset that you can use for all your future requirements reports, analytics, or business intelligence.

5 Challenges of Data Ingestion

Despite its advantages, data ingestion presents several challenges:

5.1 Maintaining Data Quality

Maintaining data quality is often challenging, especially when you have a significant amount of data. Sometimes, it can get damaged in the process of extraction or transformation. That is why its recommended you perform a data quality check regularly.

5.2 Syncing Data from Multiple Sources

Some difficulties may arise if you try to extract data from too many sources simultaneously. For example, the process might take longer to complete, or some data quality might be compromised. You must be extremely careful when trying to sync data from multiple sources simultaneously.

5.3 Streaming Data Ingestion

When dealing with real-time data processing, you must realize that a lot of information is going into the tool and waiting to be processed. The sheer amount of data might make the

process difficult for the tool and cause quality problems or slow ingestion. You will either have to limit the number of sources you use or try a better tool to deal with such data.

6 Data Ingestion Best Practices

Businesses must have accurate, up-to-date information to base their decisions, and data ingestion tools or features are perfect for that. Below are some best practices we recommend when dealing with data ingestion.

6.1 Implement Alerts at the Source

One of the best things to do is implement issue alerts at the source so that you can catch all problems early and keep them from causing more significant challenges. Remember that you can set alerts at various points, not just at the beginning.

You can set several types of data alerts, including alerts for quality, security, and availability issues. Quality alerts can tell you if there are any issues with the data quality, meaning if it is incorrect or invalid for any given reason. Security alerts can alert you to security breaches, whereas availability alerts can tell you if the data can be reached or if there are some transmission issues.

This practice aims to set at least some kind of alert to avoid having issues with your data and to keep the faulty data from compromising the quality of the rest of the batch.

6.2 Make a Copy of Raw Data

Sometimes, obtaining some data can be difficult, time-consuming, or even expensive. Even if you can get the data relatively quickly and easily, you should still be careful how you use it so you don't need to extract it again.

That is why making copies of all your raw data is important. You can use it for future references or in case some issues arise during the transformation process.

6.3 Implement Automation

As mentioned above, you should not spend time collecting data but rather implement automation and let it gather all the needed data. That is what data ingestion tools or applications are there for.

They usually come with a few simple features that do all the work. Some of these features include the following:

- Data connections: connect the application with the source documents or reports.

- Optical Character Recognition (OCR) or or OCR, is essential to extract information from various documents.
- Data wrangling will clean and format the data, transforming it into raw data you can use for any purpose.
- Data validation is a good feature if you want to check whether the collected data is accurate and up to date.
- Data processing and storage can help you move all the exported data from the data ingestion pipeline into any storage you like, whether a data warehouse, data lake, or something else.

6.4 Make Use of AI

AI can detect anomalies, process language and images, and improve ingestion accuracy when combined with ingestion tools.

7 Data Ingestion Tools and Features

As we mentioned, data ingestion tools are used to facilitate the collection and transfer of data to a target system. In most cases, the source system will have a different way of processing and storing the data, which is why choosing a good data ingestion tool is crucial.

7.1 Types of Data Ingestion Tools

If you look up data ingestion tools, you will find that quite a few are currently available online. From Rivery to Hevo, Apache, Talend, and others, these tools can be divided into four main types:

- **Hand Coding** This is a type of tool that requires a person to write the code that would help ingest the data. This practice can be time-consuming and require coding knowledge, so it might not be the best option for everyone.
- **Single-Purpose Tools** This type of tool does not require coding but allows you to use a simple interface with pre-build options to make data ingestion easier. These tools usually involve dragging and dropping.
- **Data Integration Platforms** This type of tool often needs specific integration into a domain, so you might need the help of developers to integrate the platform on your site. These tools are a bit more challenging to use and are also known to be expensive.

- **DataOps Tools** Its a type of tool that helps automate much of the process, but there is still the need for an engineer to overlook the work of the tool. Like data integration platforms, they are not as convenient as single-purpose tools.

7.2 Key Features of Data Ingestion Tools

After you select your preferred type of data ingestion tool, you should look into its features to ensure it comes with everything you need to collect the necessary data. Essential features include extraction, processing, and transformation, as well as:As weve mentioned, data ingestion involves data extraction, processing, and transformationthree essential features your tool must have.

Additionally, there are a few other features you can benefit from, such as the following:

- **Security:**You want the extracted data to be safe and secure, so ensure your tool has some protective protocol encryption.
- **Volume:** It is also essential to ensure the tool is scalable, meaning it can deal with larger volumes of information without causing significant issues.
- **Data flow tracking:** This feature lets you see how the data flows through the system. Various data ingestion tools come with different features. Which features you get depends on your chosen tool.

Various data ingestion tools come with different features. Which features you get depends on your chosen tool.

7.3 Alignment of Data Ingestion Tools with Batch and Real-Time Processing

Data ingestion tools can be broadly classified based on the mode in which they ingest data, namely batch ingestion and real-time (streaming) ingestion. Table 1 presents an alignment of commonly used data ingestion tools with their supported ingestion modes.

7.4 Mapping Tools to Key Features

Among the various data ingestion tools available, Apache Kafka is one of the most widely adopted solutions for real-time data ingestion due to its scalability, fault tolerance, and high throughput. However, Kafka requires significant operational expertise and lacks native transformation capabilities. Single-purpose tools such as Hevo and Fivetran emphasize ease of use and fast deployment but offer limited customization and can become costly at scale. Data integration platforms like Talend and Informatica provide rich transformation, security, and governance features, making them suitable for enterprise environments, although

Table 1: Alignment of Data Ingestion Tools with Batch and Real-Time Ingestion

Tool	Batch Ingestion	Real-Time Ingestion	Typical Use Case
Custom Scripts (Python, Shell)	Yes	Limited	Periodic file-based data loading
Hevo Data	Yes	Yes	Cloud-based ELT pipelines
Talend Data Integration	Yes	Limited	Enterprise ETL workflows
Apache Kafka	No	Yes	High-throughput event streaming
Apache Pulsar	No	Yes	Distributed streaming pipelines
Azure Data Factory	Yes	Limited	Cloud batch integration

Table 2: Mapping of Data Ingestion Tools to Key Features

Tool	Security	Scalability (Volume)	Data Flow Tracking
Custom Scripts	Manual	Limited	Minimal
Hevo / Fivetran	High (built-in)	Medium-High	Moderate
Talend / Informatica	Very High	High	Advanced
Apache Kafka	High (SSL, ACLs)	Very High	Strong (offsets, logs)

they involve higher complexity and licensing costs. Hand coding approaches offer maximum flexibility but demand strong programming skills and careful maintenance.

8 Challenges with Big Data Sets

We already mentioned some data ingestion challenges, including maintaining data quality, syncing data from multiple sources, and streaming ingestion. While these are common data ingestion challenges, there are a few others you should also be aware of.

For example, time efficiency can be a problem that arises if you choose data ingestion tools with hand coding. Then, limited-volume tools can cause problems with the volume of data you want to process. Furthermore, you could deal with data loss, duplicate data, and other similar issues.

If you encounter any such issue, you must try and eliminate it as soon as possible. Here are three things you should try if you want to remove all possibility of data ingestion problems:

To mitigate these issues:

1. Use automated ingestion tools: First, use a fully automated data ingestion tool that

Table 3: Alignment of Data Ingestion Tools with Batch and Real-Time Processing

Tool Category	Tool Name	Ingestion Mode	Remarks
Hand Coding	Custom Python / Shell Scripts	Batch	High flexibility; suitable for scheduled file-based ingestion but limited scalability and monitoring.
Single-Purpose Tools	Hevo Data	Batch + Near Real-Time	No-code cloud platform supporting automated pipelines with built-in security and scalability.
Data Integration Platforms	Talend Data Integration	Batch (Primarily)	Enterprise-grade ETL tool with strong transformation, governance, and metadata management.
DataOps / Streaming Tools	Apache Kafka	Real-Time	Distributed event-streaming platform optimized for high-throughput, fault-tolerant real-time ingestion.
DataOps / Streaming Tools	Apache Pulsar	Real-Time	Cloud-native messaging system with multi-tenancy and low-latency streaming support.
Cloud Integration Tools	Azure Data Factory	Batch (Limited Streaming)	Managed cloud service for batch-oriented data movement and orchestration.

eliminates the possibility of human error since the tool itself will do the work for you. However, remember not to go for the first tool you come across. Research many tools thoroughly and go for a trusted, well-established tool with many satisfied customers

2. **Implement data SLAs(Service Level Agreement):**Second, implement data SLAs to help you learn more about what your customers expect from your business, what they think you can improve, etc..
3. **Quality Checks:** Third, do quality checks often to ensure the data collected has no issues. As mentioned above, you can implement alerts at the source to get notified as soon as the tool detects some problem.

9 Data Ingestion vs. ETL

Data ingestion focuses on collecting and storing data for flexible use, supporting real-time and batch processing. ETL (Extract, Transform, Load) transforms data before storage and typically operates on batch schedules.

While ETL remains in use, data ingestion and ELT are more common in modern architectures. The choice depends on business requirements and data strategy.

10 Tips for Choosing the Right Data Ingestion Tool

When selecting a tool:

- Choose one that improves productivity and business insights.
- Ensure it solves key business challenges.
- Prefer tools that automate the ingestion process.

11 Some Question

Multiple-Choice Questions (MCQs)

1. What is data ingestion?
 - A. The process of storing data permanently in a database
 - B. The process of collecting and importing data from multiple sources for analysis
 - C. The process of visualizing data
 - D. The process of deleting outdated data
2. How does data ingestion differ from ETL?
 - A. Data ingestion transforms data before loading, ETL does not
 - B. Data ingestion and ETL are exactly the same
 - C. ETL transforms data before loading, while data ingestion may or may not
 - D. ETL supports real-time data, data ingestion does not
3. Which of the following are the two main types of data ingestion?
 - A. Manual and automated
 - B. Structured and unstructured

- C. Batch and real-time
 - D. Centralized and decentralized
4. What is the primary purpose of a data ingestion pipeline?
- A. To visualize data
 - B. To move data from source systems to a storage or processing system
 - C. To delete duplicate records
 - D. To archive historical data only
5. Which of the following is a commonly used data ingestion tool?
- A. Microsoft Word
 - B. Apache Kafka
 - C. Adobe Photoshop
 - D. PowerPoint

Short-Answer Questions

1. Define data ingestion.
2. Is data ingestion the same as ETL? Briefly explain.
3. Name the two main types of data ingestion.
4. What is a data ingestion pipeline?
5. Give one example of a data ingestion tool.

Long-Answer / Essay Questions

1. Explain data ingestion and discuss its importance in modern data systems.
2. Compare data ingestion and ETL, highlighting their key differences.
3. Describe the types and methods of data ingestion with examples.

Data Ingestion: Concepts, Benefits, Challenges, and Tools

ETL vs ELT: Whats the Difference?

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1 Introduction

ETL and **ELT** are two common approaches in data integration. Their main task is to transfer data from one place to another. However, each has unique characteristics and is suitable for different data needs.

The main difference between ETL and ELT is the *order of operations*. **ETL (Extract, Transform, Load)** transforms data before loading it into the target system. **ELT (Extract, Load, Transform)** loads raw data first, then transforms it within the destination system, typically using cloud-based data warehouses.

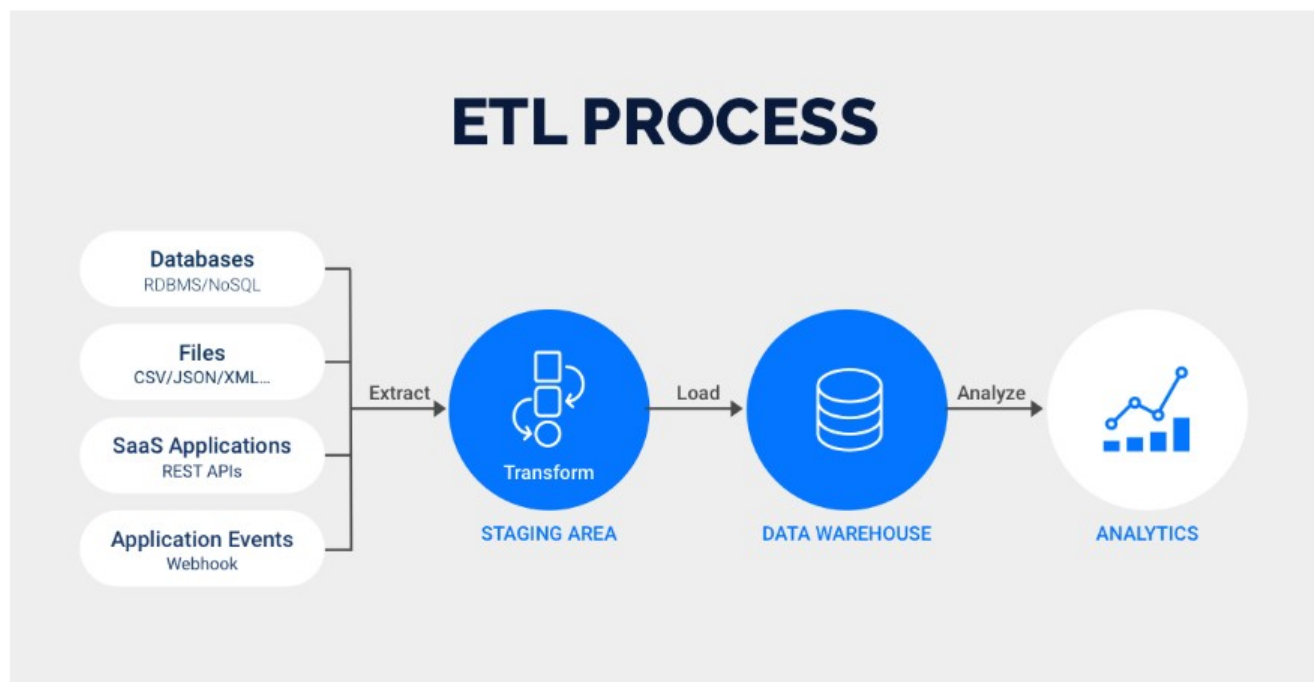
ETL involves transforming data on a separate processing server before transferring it to the data warehouse. **ELT**, on the other hand, performs data transformations directly within the data warehouse itself, allowing raw data to be sent directly to the destination system.

ETL is an older method ideal for complex transformations of smaller data sets and for organizations prioritizing data security. ELT is a newer technology that provides more flexibility to analysts and is well suited for both structured and unstructured data.

Your decision between ETL and ELT will determine your data storage, analysis, and processing strategies. Before choosing between the two methods, it is important to consider your business type and data needs.

2 What is ETL (Extract, Transform, Load)?

Extract, Transform, and Load (ETL) is a data integration methodology that extracts raw data from source systems, transforms the data on a secondary processing server, and then loads the transformed data into a target database.

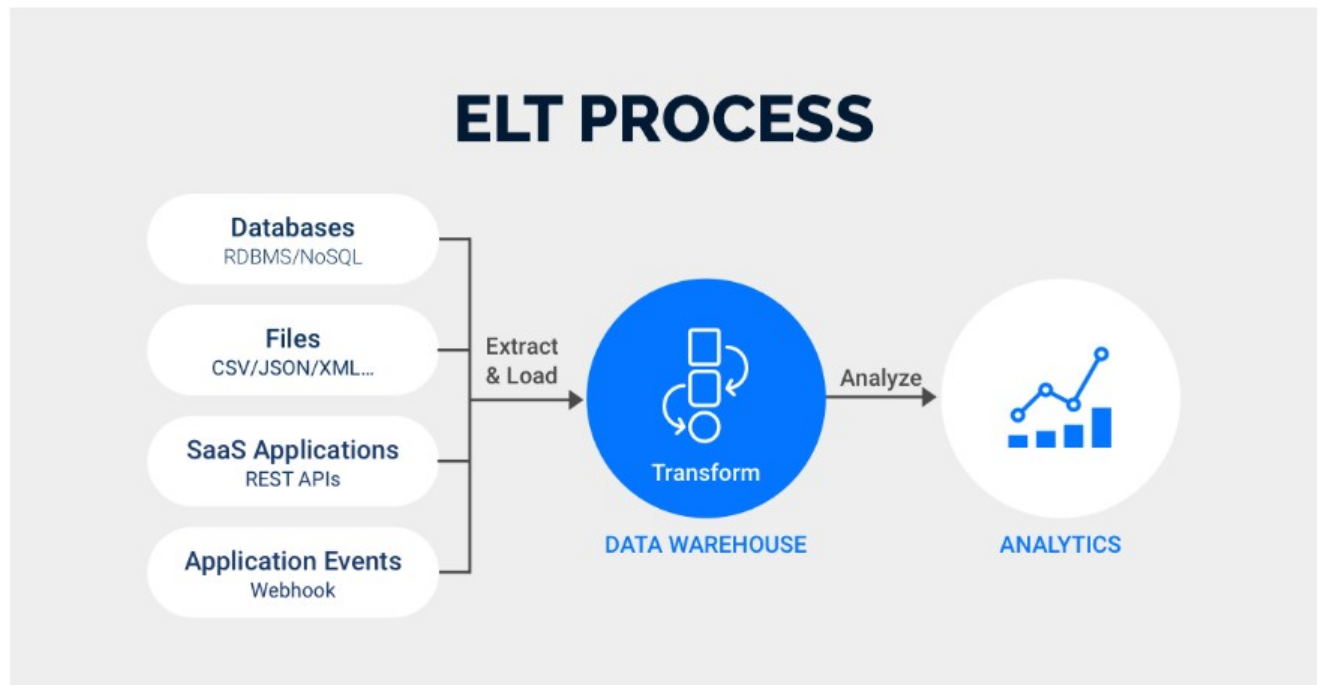


ETL is commonly used when data must be transformed to conform to the schema or constraints of a target database. The method emerged in the 1970s and remains prevalent among on-premise databases with limited memory and processing power.

For example, Online Analytical Processing (OLAP) data warehouses typically accept only relational SQL-based data structures. ETL ensures compliance by routing extracted data to a processing server, transforming non-conforming data into SQL-based formats, and loading it into the warehouse only after successful transformation.

3 What is ELT (Extract, Load, Transform)?

Unlike ETL, Extract, Load, and Transform (ELT) does not require data transformations prior to loading. ELT loads raw data directly into a target data warehouse, where cleansing, enrichment, and transformation occur internally. Raw data is stored indefinitely in the data



warehouse, enabling multiple transformations over time. ELT became possible due to the rise of scalable cloud-based data warehouses.

Platforms such as Snowflake, Amazon Redshift, Google BigQuery, and Microsoft Azure provide sufficient storage and processing power to support raw data repositories and in-database transformations.

Although ELT is not universally adopted, it continues to gain popularity as organizations migrate to cloud infrastructure.

4 How is ETL Different from the ELT Process?

ETL and ELT differ in two primary ways. One difference is where the data is transformed, and the other difference is how data warehouses retain data.

1. ETL transforms data on a separate processing server, while ELT transforms data within the data warehouse itself.
2. ETL does not transfer raw data into the data warehouse, while ELT sends raw data directly to the data warehouse.

ELT processes data faster than ETL. ETL includes a preliminary transformation step before loading data into the target, which becomes difficult to scale and slows performance as data size grows. ELT, in contrast, loads data directly into the target system, transforming it in parallel.

For ETL, the **process of data ingestion** is made slower by transforming data on a separate server before the loading process.

ELT, in contrast, delivers faster data ingestion, because data is not sent to a secondary server for restructuring. In fact, with ELT, data can be loaded and transformed simultaneously.

The raw data retention of ELT creates a rich historical archive for generating business intelligence. As goals and strategies change, BI teams can re-query raw data to develop new transformations using comprehensive datasets. ETL, on the other hand, does not generate complete raw data sets that are endlessly queryable.

These factors make ELT more flexible, efficient, and scalable, especially for ingesting large amounts of data, processing data sets that contain both structured and unstructured data, and developing diverse business intelligence.

How you process unstructured data is crucial. At the moment, ELT is the best option for it. It provides superior processing of semi-structured and unstructured data compared to ETL, which is typically used for structured data.

The majority of data is unstructured (images, videos, PDF files, PowerPoint documents, etc.), yet this type of data is still harder to access and process. In the future, the industry will focus on eliminating difficulties and improving the interpretation of this unstructured data, and ELT will play a significant role in that.

On the other hand, ETL is ideal for compute-intensive transformations, systems with legacy architectures, or data workflows that require manipulation before entering a target system, such as erasing personal identifying information (PII).

The data pipeline of both ETL and ELT includes cleaning and filtering. It is a key part of the data transformation process. And because the ETL method completes transformation before loading data into the server, it is better for meeting compliance standards and transferring sensitive data.

Most companies are required to encrypt, remove, or mask data to protect their clients privacy. If they do not try their best to do so, they may violate compliance standards and risk the clients sensitive data.

Sometimes this happens unintentionally. But with ETL, you will reduce the risk of transferring non-compliant data. Why? Because of the data pipeline, meaning the data is cleaned and filtered before it leaves its initial destination.

ELT is generally more flexible, scalable, and efficient for large datasets, unstructured data, and advanced analytics. ETL remains preferable for compute-intensive transformations, legacy systems, or workflows requiring strict pre-load data compliance, such as removing

personally identifiable information (PII).

Category	ETL	ELT
Definition	Data is transformed before loading into the destination system.	Data is loaded first, then transformed inside the destination system.
Extract	Raw data extracted via API connectors.	Raw data extracted via API connectors.
Transform	Transformation occurs on a secondary server.	Transformation occurs inside the target system.
Load	Transformed data is loaded into destination.	Raw data is loaded directly.
Speed	Slower due to pre-load transformations.	Faster with parallel in-database processing.
Transformations	Best for compute-intensive pre-processing.	In-database, scalable transformations.
Maturity	Established for over 20 years.	Newer methodology.
Privacy	Pre-load transformations help remove PII.	Requires stronger privacy safeguards.
Maintenance	Requires additional servers.	Reduced maintenance overhead.
Costs	Higher due to separate infrastructure.	Lower due to simplified architecture.
Requeries	Raw data not retained.	Raw data can be queried indefinitely.
Data Lake Compatibility	No	Yes
Data Output	Structured data.	Structured, semi-structured, unstructured.
Data Volume	Best for small datasets.	Ideal for large datasets.

5 A Brief History of ETL and ELT

Today, businesses across the spectrum require data integration to centralize, access, and activate data across their organizations. Businesses must leverage dozens or hundreds of different data sources, across countries, continents, and teams, to drive results in the data-driven economy. In this complex, fractured landscape, combining multiple data sources into a unified view has never been more important. But this challenge is nothing new. Data

integration has vexed organizations since the dawn of the digital era.

The first inventions of the modern computer age laid the groundwork for the unique function of data integration. In the late 1960s, disk storage supplanted punch cards, enabling direct access to data. Not long after, IBM and other companies pioneered the first Database Management Systems (DBMS). These advancements soon led to the sharing of data between computers. Almost immediately, the cumbersome process of integrating data and data sources with external machines became a challenge.

ETL, the first standardized method for facilitating data integration, emerged in the 1970s. ETL rose to prominence as enterprise businesses adopted multi-pronged computer systems and heterogeneous data sources. These businesses needed a way to aggregate and centralize data from transactions, payroll systems, inventory logs, and other enterprise resource planning (ERP) data.

With the rise of data warehouses in the 1980s, ETL became even more essential. Data warehouses could integrate data from a variety of sources, but typically required custom ETLs for each data source. This led to an explosion of **ETL tools**. By the end of the 1990s, many of these solutions finally became affordable and scalable for mid-market businesses, not just large enterprises.

With the emergence of cloud computing in the 2000s, cloud data lakes and data warehouses caused a new evolution: ELT. With ELT, businesses could load unlimited raw data straight into a cloud DWH. Engineers and analysts could execute an infinite number of SQL queries on top of this raw data, directly inside the cloud data warehouse itself.

For the first time, businesses could unlock the analytical firepower and efficiency that big data had always promised. Combined with visualization tools and cloud DWHs, ELTs ushered in a new era of analytics and data-driven decision-making.

6 Which is Better: ETL or ELT?

Today, businesses across the spectrum require data integration to centralize, access, and activate data across their organizations. Businesses must leverage dozens or hundreds of different data sources, across countries, continents, and teams, to drive results in the data-driven economy. In this complex, fractured landscape, combining multiple data sources into a unified view has never been more important. But this challenge is nothing new. Data integration has vexed organizations since the dawn of the digital era.

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For the first time, businesses could unlock the analytical firepower and efficiency that big data had always promised. Combined with visualization tools and cloud DWHs, ELTs ushered in a new era of analytics and data-driven decision-making.

ELT is building popularity due to its scalability and speed, particularly for large data sets and real-time processing. However, ETL is valuable for complex transformations and legacy systems.

The choice between ETL and ELT processes depends on your specific data requirements, with both methods coexisting to handle different needs. Therefore, it's challenging to say whether ELT will replace ETL.

7 Some Questions

1. What is the difference between ETL and ELT?
2. When should I use ETL and when should I use ELT?
3. Are there any performance differences between ETL and ELT?

Data Ingestion: Concepts, Benefits, Challenges, and Tools

Introduction to Apache Airflow

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February 18, 2026

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1 Introduction

In the modern era of data-driven decision-making, organizations process enormous volumes of data daily. Managing data pipelines, orchestrating workflows, and ensuring reliable execution of complex dependencies require robust orchestration platforms. Apache Airflow has emerged as one of the most widely adopted open-source workflow orchestration tools in the data engineering ecosystem.

Originally developed at Airbnb and later donated to the Apache Software Foundation, Airflow enables developers and data engineers to define workflows programmatically using Python. Unlike traditional ETL tools with rigid graphical interfaces, Airflow embraces the concept of "configuration as code," making workflows version-controllable, testable, and modular.

This paper provides a comprehensive exploration of Apache Airflow, covering its architecture, core concepts, execution model, scalability, deployment strategies, monitoring, security, and best practices in production environments.

2 Architecture Overview

Apache Airflow follows a modular and distributed architecture designed for scalability and reliability.

2.1 Core Components

2.1.1 Scheduler

The scheduler is the central orchestration engine of Airflow. It:

- Parses DAG files
- Determines task execution times
- Checks dependencies
- Queues tasks for execution

The scheduler continuously monitors DAG definitions and the metadata database to determine which tasks are ready for execution.

2.1.2 Metadata Database

Airflow stores its operational metadata in a relational database such as PostgreSQL or MySQL. This database stores:

- DAG definitions metadata
- Task states
- Execution logs metadata
- User authentication information

Separation of metadata storage enables horizontal scalability and production-grade deployments.

2.1.3 Executor

The executor determines how tasks are executed. Airflow supports multiple executors:

- **SequentialExecutor** Single-threaded, suitable for development.
- **LocalExecutor** Parallel execution on a single machine.
- **CeleryExecutor** Distributed execution using Celery workers.
- **KubernetesExecutor** Executes tasks as Kubernetes pods.

2.1.4 Web Server

The Airflow Web UI provides:

- DAG visualization
- Task logs
- Manual trigger capabilities
- Monitoring dashboards

2.1.5 Workers

In distributed configurations (Celery or Kubernetes), workers execute tasks in parallel, improving performance and throughput.

3 Key Concepts

3.1 DAGs (Directed Acyclic Graphs)

A DAG represents a workflow as a graph of tasks with dependencies. DAGs must be acyclic, meaning no circular dependencies are allowed.

3.2 Tasks

Tasks are units of execution derived from operators. Each task instance corresponds to a specific execution date.

3.3 Operators

Operators define the actual work performed. Examples include:

- BashOperator
- PythonOperator
- EmailOperator
- PostgresOperator

3.4 Dependencies

Dependencies define execution order using:

```
task1 >> task2
task3.set_upstream(task2)
```

3.5 Sensors

Sensors wait for conditions such as:

- File availability
- External task completion
- Database record existence

3.6 Hooks

Hooks abstract connections to external systems such as:

- Databases
- Cloud services
- APIs

4 Execution Model

Airflow follows a pull-based execution model:

1. Scheduler identifies ready tasks.
2. Tasks are queued.
3. Executor assigns tasks to workers.
4. Workers execute tasks.
5. State updates are recorded in metadata DB.

Each task instance transitions through states:

- scheduled
- queued

- running
- success
- failed
- retry

5 Dynamic Workflow Generation

Airflow allows dynamic DAG creation using Python logic.

```
for i in range(5):
    PythonOperator(
        task_id=f"task_{i}",
        python_callable=my_function,
        dag=dag
    )
```

Dynamic generation enables:

- Parameterized pipelines
- Multi-tenant workflows
- Metadata-driven DAG construction

6 Example DAG

```
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime

def extract():
    print("Extracting data")

def transform():
    print("Transforming data")

def load():
    print("Loading data")
```

```

with DAG(
    dag_id="etl_pipeline",
    start_date=datetime(2024,1,1),
    schedule_interval="@daily",
    catchup=False
) as dag:

    task1 = PythonOperator(
        task_id="extract",
        python_callable=extract
    )

    task2 = PythonOperator(
        task_id="transform",
        python_callable=transform
    )

    task3 = PythonOperator(
        task_id="load",
        python_callable=load
    )

    task1 >> task2 >> task3

```

7 Scalability and Distributed Execution

Airflow scales through:

- Distributed workers
- Horizontal scaling of scheduler
- Kubernetes pod-based execution

Large enterprises often deploy:

- Multiple schedulers
- Highly available metadata databases
- Redis/RabbitMQ brokers for Celery

8 Monitoring and Observability

Monitoring includes:

- Task logs
- SLA monitoring
- Email alerts
- Integration with Prometheus and Grafana

Airflow supports:

- Retry mechanisms
- Failure notifications
- Task duration tracking

9 Security Considerations

Security features include:

- Role-Based Access Control (RBAC)
- LDAP/OAuth integration
- Encrypted connections
- Secrets backend integration

Sensitive credentials can be stored using:

- HashiCorp Vault
- AWS Secrets Manager
- Environment variables

10 Best Practices in Production

- Keep DAG files lightweight
- Avoid heavy imports at global scope
- Use retries wisely
- Implement idempotent tasks
- Use proper logging
- Separate development and production environments

11 Use Cases

Airflow is widely used for:

- ETL/ELT pipelines
- Machine learning workflows
- Data warehouse orchestration
- Batch processing
- API orchestration

12 Comparison with Other Tools

Tool	Strength	Weakness
Airflow	Python-based, flexible	Requires engineering expertise
Luigi	Lightweight	Limited UI
Prefect	Modern orchestration	Smaller ecosystem

13 Deployment Strategies

Airflow can be deployed:

- On-premises
- Docker-based

- Kubernetes cluster
- Managed services (MWAA, Composer)

Production deployments require:

- Load balancer
- External database
- Message broker
- Backup strategy

14 Conclusion

Apache Airflow has become a cornerstone technology in modern data engineering. Its flexibility, scalability, and strong open-source community make it a powerful orchestration solution. While it requires engineering discipline and operational maturity, it provides unmatched control over complex workflows.

As organizations continue to embrace data-driven architectures, workflow orchestration platforms like Airflow will remain critical infrastructure components enabling scalable, reliable, and maintainable data ecosystems.

15 Questions

Explain DAG scheduling in Airflow.

Difference between LocalExecutor and CeleryExecutor?

What causes scheduler bottlenecks?

How does Airflow scale horizontally?

What is idempotent task design?

How would you deploy this in Kubernetes?